

AP CALCULUS AB – UNIT 1

Limits and Continuity

Comprehensive Coloured Review



Name: _____ Class: _____ Date: _____

Review Scope and Learning Objectives

- Interpret and estimate limits from symbolic, graphical, and numerical representations.
- Evaluate limits using limit laws, algebraic manipulation, and the Squeeze Theorem.
- Analyze continuity, classify discontinuities, and repair removable discontinuities.
- Connect infinite and end-behavior limits with asymptotes and apply the Intermediate Value Theorem.

Website Lesson Map

Lesson	Website title	Review target
1.1	Can Change Occur at an Instant?	Master the lesson objective and connect it to limits and continuity.
1.2	Defining Limits and Using Limit Notation	Master the lesson objective and connect it to limits and continuity.
1.3	Estimating Limit Values from Graphs	Master the lesson objective and connect it to limits and continuity.
1.4	Estimating Limit Values from Tables	Master the lesson objective and connect it to limits and continuity.
1.5	Determining Limits Using Algebraic Properties of Limits	Master the lesson objective and connect it to limits and continuity.
1.6	Determining Limits Using Algebraic Manipulation	Master the lesson objective and connect it to limits and continuity.
1.7	Selecting Procedures for Determining Limits	Master the lesson objective and connect it to limits and continuity.
1.8	Determining Limits Using the Squeeze Theorem	Master the lesson objective and connect it to limits and continuity.
1.9	Connecting Multiple Representations of Limits	Master the lesson objective and connect it to limits and continuity.
1.10	Exploring Types of Discontinuities	Master the lesson objective and connect it to limits and continuity.
1.11	Defining Continuity at a Point	Master the lesson objective and connect it to limits and continuity.
1.12	Confirming Continuity over an Interval	Master the lesson objective and connect it to limits and continuity.
1.13	Removing Discontinuities	Master the lesson objective and connect it to limits and continuity.
1.14	Connecting Infinite Limits and Vertical Asymptotes	Master the lesson objective and connect it to limits and continuity.
1.15	Connecting Limits at Infinity and Horizontal Asymptotes	Master the lesson objective and connect it to limits and continuity.

1.16

Working with the Intermediate Value Theorem

Master the lesson objective and connect it to limits and continuity.

Key Knowledge, Formulas, and Theorems

- Two-sided existence: $\lim_{x \rightarrow c} f(x) = L$ only when the left- and right-hand limits both equal L .
- Continuity at c : $f(c)$ exists, $\lim_{x \rightarrow c} f(x)$ exists, and the two values are equal.
- For rational functions, factor or rationalize before canceling; cancellation describes nearby behavior, not the missing point.
- Infinite limits describe unbounded output near a finite input; limits at infinity describe end behavior.
- IVT requires continuity on a closed interval and guarantees existence, not uniqueness or an exact location.

Strategy and Precision Checklist

1. Identify the representation and the exact quantity requested before calculating.
2. Write the relevant definition, theorem, or derivative rule before substitution.
3. Preserve exact values in non-calculator work; use radian mode and appropriate rounding in calculator work.
4. For contextual conclusions, include the input, sign, units, and meaning.
5. Justifications must state why a theorem or procedure applies, not merely name it.

Solved Mixed Examples**Solved Example: Example 1 – 1.6**

Evaluate $\lim_{x \rightarrow 3} \frac{\sqrt{x} - \sqrt{3}}{x - 3}$.

Answer and Reasoning

Result: $\frac{1}{2\sqrt{3}}$. Multiply by the conjugate. The quotient becomes $1/(\sqrt{x} + \sqrt{3})$ for nearby x , then substitute.

Solved Example: Example 2 – 1.8

If $|g(x)| \leq 2|x - 2|$ near $x = 2$, then $\lim_{x \rightarrow 2} g(x)$ equals

Answer and Reasoning

Result: 0. Both bounding functions approach 0, so the Squeeze Theorem forces $g(x)$ to approach 0.

Solved Example: Example 3 – 1.13

For what value of k is $f(x) = \begin{cases} kx + 1, & x < 2 \\ x^2 + 4, & x \geq 2 \end{cases}$ continuous at $x = 2$?

Answer and Reasoning

Result: $\frac{7}{2}$. Continuity requires $k(2) + 1 = 8$.

Solved Example: Example 4 – 1.14

The function $f(x) = \frac{x+1}{(x-3)^2}$ has which behavior as $x \rightarrow 3$?

Answer and Reasoning

Result: $f(x) \rightarrow +\infty$ from both sides. The squared denominator is positive and approaches 0, while the numerator keeps a fixed sign.

Leveled Mixed Practice**Foundation**

Questions in this band integrate the website lessons at an increasingly demanding level.

1. Suppose $\lim_{x \rightarrow -1^-} f(x) = -4$ and $\lim_{x \rightarrow -1^+} f(x) = -2$. Which statement is correct?

2. Evaluate $\lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$.

3. Evaluate $\lim_{x \rightarrow 3} \frac{\sqrt{x} - \sqrt{3}}{x - 3}$.

4. If $|g(x)| \leq 2|x-2|$ near $x = 2$, then $\lim_{x \rightarrow 2} g(x)$ equals

Core

Questions in this band integrate the website lessons at an increasingly demanding level.

5. For what value of k is $f(x) = \begin{cases} kx + 1, & x < 2 \\ x^2 + 4, & x \geq 2 \end{cases}$ continuous at $x = 2$?

6. The function $f(x) = \frac{x+1}{(x-3)^2}$ has which behavior as $x \rightarrow 3$?

7. Find $\lim_{x \rightarrow \infty} \frac{5x^3 + 1}{2x^3 - 2}$.

8. A function f is continuous on $[0, 2]$, with $f(0) = -4$ and $f(2) = 5$. Which conclusion is guaranteed?

Advanced

Questions in this band integrate the website lessons at an increasingly demanding level.

9. Which discontinuity can be removed by redefining a single function value?

10. If f is continuous at $x = 2$ and $\lim_{x \rightarrow 2} f(x) = 7$, then which must be true?

11. Suppose $\lim_{x \rightarrow 3^-} f(x) = 1$ and $\lim_{x \rightarrow 3^+} f(x) = 1$. Which statement is correct?

12. Evaluate $\lim_{x \rightarrow -3} \frac{(x - -3)(x + 5)}{x - -3}$.

Difficult

Questions in this band integrate the website lessons at an increasingly demanding level.

13. Evaluate $\lim_{x \rightarrow 1} \frac{\sqrt{x} - \sqrt{1}}{x - 1}$.

14. If $|g(x)| \leq 4|x - 2|$ near $x = 2$, then $\lim_{x \rightarrow 2} g(x)$ equals

15. Use numerical evidence to determine the sign of $\lim_{x \rightarrow 1^-} \frac{1}{x-1}$.

16. A calculator table near $x = 1$ gives $f(0.9000)=3.2090$; $f(0.9900)=3.3590$; $f(0.9990)=3.3423$; $f(0.9999)=3.2840$. Which is the best estimate of $\lim_{x \rightarrow 1^-} f(x)$?

Challenge

Questions in this band integrate the website lessons at an increasingly demanding level.

17. Use a graphing calculator to estimate $\lim_{x \rightarrow 2.34} \frac{\sin(x-2.34)}{x-2.34}$.

18. A graph of $f(x) = \frac{x^2-9}{x-3}$ shows a hole. What is the y-coordinate of the hole?

19. Use numerical evidence to determine the sign of $\lim_{x \rightarrow 1^-} \frac{1}{x-1}$.

20. A calculator table near $x = 1$ gives $f(0.9000)=1.5180$; $f(0.9900)=1.6680$; $f(0.9990)=1.5847$; $f(0.9999)=1.6430$. Which is the best estimate of $\lim_{x \rightarrow 1^-} f(x)$?

AP-Style Mixed Checkpoint

1. Suppose $\lim_{x \rightarrow 1^-} f(x) = 5$ and $\lim_{x \rightarrow 1^+} f(x) = 5$. Which statement is correct?

- (A) $\lim_{x \rightarrow 1} f(x) = 0$ (B) $f(1) = 5$ must hold
(C) $\lim_{x \rightarrow 1} f(x) = 5$ (D) The limit does not exist

2. Evaluate $\lim_{x \rightarrow 1} \frac{(x-1)(x+4)}{x-1}$.

- (A) 0 (B) 5
(C) The limit does not exist (D) 1

3. Evaluate $\lim_{x \rightarrow 3} \frac{\sqrt{x} - \sqrt{3}}{x - 3}$.

- (A) The limit does not exist (B) 0
(C) $2\sqrt{3}$ (D) $\frac{1}{2\sqrt{3}}$

4. If $|g(x)| \leq 3|x - 2|$ near $x = 2$, then $\lim_{x \rightarrow 2} g(x)$ equals

- (A) It cannot be determined (B) 0
(C) 3 (D) The limit does not exist

5. For what value of k is $f(x) = \begin{cases} kx + 1, & x < 2 \\ x^2 + 2, & x \geq 2 \end{cases}$ continuous at $x = 2$?

- (A) $\frac{3}{2}$ (B) $\frac{7}{2}$
(C) 6 (D) $\frac{5}{2}$

6. The function $f(x) = \frac{x+1}{(x+2)^2}$ has which behavior as $x \rightarrow -2$?

- (A) $f(x) \rightarrow -\infty$ from both sides (B) $f(x) \rightarrow 0$
(C) There is a removable discontinuity (D) The one-sided limits are opposite infinities

7. Find $\lim_{x \rightarrow \infty} \frac{5x^3 + 1}{4x^3 - 2}$.

- (A) $+\infty$ (B) $\frac{5}{4}$
(C) 0 (D) $\frac{4}{5}$

8. A function f is continuous on $[0, 2]$, with $f(0) = -4$ and $f(2) = 5$. Which conclusion is guaranteed?

- (A) f is differentiable on $(0, 2)$ (B) There is exactly one such c
(C) There is at least one $c \in (0, 2)$ with $f(c) = 1$ (D) f is increasing on $[0, 2]$

Free-Response Synthesis

FRQ1 – Limits from numerical data

A table gives $f(1.9) = 3.80$, $f(1.99) = 3.98$, $f(2.01) = 4.02$, and $f(2.1) = 4.20$, while $f(2) = 7$.

(a) Estimate the left- and right-hand limits and state whether the two-sided limit exists. [3 points]

(b) Explain why the value $f(2) = 7$ does not change the limit. [2 points]

(c) Redefine $f(2)$ so that the function is continuous at $x = 2$. [2 points]

FRQ2 – Algebraic limit and removable discontinuity

Let $g(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$.

(a) Evaluate $\lim_{x \rightarrow 3} g(x)$ and justify the algebraic step. [3 points]

(b) Define $g(3)$ so that the extended function is continuous. [1 points]

(c) Describe the graph before and after the redefinition. [2 points]

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